

capabilities are operating as intended, functionality testing is done. Testing user interfaces, data input/output, calculations, and particular water supply management features are all part of this process. Performance testing assesses the application's responsiveness and effectiveness. It assesses how quickly the application responds to various actions, looks for potential performance bottlenecks, and determines whether it can scale to handle a large volume of users or data. To evaluate an application's usability, usability testing entails getting feedback from actual users or testers. This aids in detecting any problems with the user interface, directions, navigation, or general user experience. The input gathered aids in boosting user happiness and application design refinement. Testing data integrity and correctness is essential to ensuring that the application handles data correctly. Cross-referencing data inputs with known values, validating data calculations or algorithms, and guaranteeing data consistency across the application constitute every part of this process.

3.4 Prediction

In a water supply application, prediction entails looking ahead to potential water supply situations using historical data and modelling methods. The main steps in the prediction process are as follows: First, historical statistics on the availability of water, including information on water use, rainfall, temperature, population, and other pertinent factors, are acquired. For the purpose of capturing seasonal variations and long-term trends, this data should span a sizable time period. The acquired data is then put through preprocessing, which involves managing missing numbers, outliers, and inconsistencies to clean it up. To prepare the data for modelling, procedures including feature engineering, normalisation, and data transformation may be used. Feature selection is then conducted to identify the most relevant features that impact water supply, such as historical consumption patterns, weather conditions, or demographic factors.